

Introduction to AI Safety, Ethics, and Society — Ch1 Quiz

ANS

1. In the course's taxonomy of major sources of AI risk, which category covers risks from actors deliberately using AI systems to cause harm?

- A) AI Race
- B) Organizational Risks
- C) Malicious Use
- D) Rogue AIs

ANS: C) Malicious Use

2. Per the White House Executive Order definition referenced in the slides, a “dual-use foundation model” contains at least how many parameters?

- A) Tens of millions
- B) Hundreds of millions
- C) Tens of billions
- D) Trillions

ANS: C) Tens of billions

3. Which of the following is NOT one of the three risk pathways the EO uses to describe how a dual-use foundation model could pose a serious risk?

- A) Substantially lowering the barrier to design or use CBRN weapons
- B) Enabling powerful offensive cyber operations
- C) Permitting evasion of human control through deception or obfuscation
- D) Increasing the energy consumption of data centers

ANS: D) Increasing the energy consumption of data centers

4. Which of the following is listed in the slides as a way to reduce malicious use of AI?

- A) Open-sourcing all frontier models by default
- B) Structured access, especially for biological uses
- C) Removing legal liability from developers
- D) Eliminating anomaly detection systems

ANS: B) Structured access, especially for biological uses

5. The slides describe the weaponization of AI in military contexts as comparable to which historical development?

- A) The invention of the printing press
- B) The “third revolution in warfare,” alongside gunpowder and nuclear arms
- C) The industrial revolution
- D) The invention of radar

ANS: B) The “third revolution in warfare,” alongside gunpowder and nuclear arms

6. What does the “Swiss Cheese model” referenced in the slides illustrate about organizational safety?

- A) A single strong defense is always sufficient

- B) Layering multiple imperfect defenses compensates for each other's weaknesses and reduces overall risk
- C) Safety measures are unnecessary with a skilled team
- D) Cheese-making inspired AI hardware design

ANS: B) Layering multiple imperfect defenses compensates for each other's weaknesses and reduces overall risk

7. In the “intelligence explosion” thought experiment, what is the projected outcome of running 10,000 copies of a world-class AI researcher at 100× human pace?

- A) Progress equivalent to a few years compressed into a decade
- B) Progress equivalent to many decades compressed into a few months
- C) No meaningful change in the pace of AI progress
- D) A slowdown in research due to coordination costs

ANS: B) Progress equivalent to many decades compressed into a few months

8. Which of the following best describes “structured access” as a risk-reduction strategy for AI systems?

- A) Publicly releasing model weights so anyone can inspect them
- B) Providing AI capabilities through controlled interfaces (e.g., APIs) rather than open weight releases
- C) Requiring all AI research to be published in peer-reviewed journals
- D) Banning commercial use of AI models entirely

ANS: B) Providing AI capabilities through controlled interfaces (e.g., APIs) rather than open weight releases

9. Why do the slides emphasize structured access “especially for biological uses”?

- A) Biological applications are the most profitable use case for AI
- B) Once weights are open-sourced or leaked, dangerous capabilities can't be revoked, unlike API-based access which can be monitored and cut off
- C) Biology-related queries require more computing power than other uses
- D) Regulators only oversee biological applications of AI

ANS: B) Once weights are open-sourced or leaked, dangerous capabilities can't be revoked, unlike API-based access which can be monitored and cut off

10. According to the slides, what is a key risk associated with automated warfare?

- A) It has no effect on the speed or accountability of military decision-making
- B) Automatic retaliation systems could escalate accidents into full-scale war, and automation could reduce accountability for military leaders
- C) It eliminates the possibility of accidental conflict entirely
- D) It only affects naval warfare, not other military domains

ANS: B) Automatic retaliation systems could escalate accidents into full-scale war, and automation could reduce accountability for military leaders